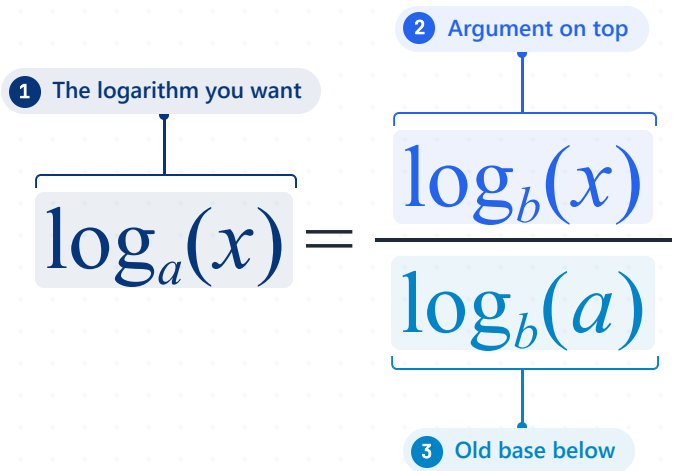
THE FORMULA, PART BY PART

Change of base

any logarithm becomes a quotient of two logarithms in a base the calculator knows



1 The logarithm you want $\log_a(x)$

Base a, argument x, in a base the calculator may not have. Example: $\log_2(10)$ asks which power of 2 gives 10. It must lie between 3 and 4, because $2^3 = 8$ and $2^4 = 16$.

2 Argument on top *numerator*

Any base works as long as top and bottom use the same one; in practice 10 or e, which a calculator has. Example: for $\log_2(10)$ in base 10, the top is $\log(10) = 1$.

3 Old base below *denominator*

Swapping top and bottom gives the reciprocal, a different number. Example: the bottom is $\log(2) \approx 0.301$, so $\log_2(10) = 1 / 0.301 \approx 3.322$. Upside down it would give 0.301, not a power between 3 and 4.